

Danil Zanozin

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EDUCATION

California State University Channel Islands

Camarillo, CA

Bachelor of Science in Computer Science, Minors in Cybersecurity and Mathematics

Sep. 2024 – May 2026

- Incoming M.S. student at CSUCI – starting **Fall 2026**

Santa Barbara City College

Santa Barbara, CA

Associate of Science in Computer Science, Associate of Arts in Mathematics

Sep. 2022 – May 2024

- **3.84 GPA**, Top 5% of class. Honoree of SBCC Foundation, **Presidents Honor Roll** Fall 2023 & Spring 2024; Holmes CS Scholarship Recipient; PTK Honor's Society Member; Competitive Programming Club Project Manager

PUBLICATIONS

PII Virtualization: Reconciling Data Privacy and Tool-Calling Utility in LLM Agents 2026

- *Sole author; IEEE-format manuscript, under review.* Reversible session-scoped tokenization mechanism for LLM tool-calling agents: two-stage regex detection, SHA-256 + Fernet-encrypted vault, **fail-closed leak guard**. Evaluated on a live production deployment: **zero tool-call degradation with zero raw PII exposure** to the external model provider – a working **LLM output containment** mechanism demonstrated under real operating conditions

Agentic AI for Autonomous Cyber Defense: Architecture, Threat Modeling & Evaluation 2026

- *Co-author with Abdoolee, R. & Vakilian, V.; IEEE-format manuscript in preparation.* Formalized agentic cybersecurity agents as **constrained POMDPs**; structured **adversarial threat model**; **NIST CSF-aligned** staged-autonomy taxonomy; evaluated on LANL auth/DNS/flows telemetry

EXPERIENCE

Undergraduate Research Assistant (Paid)

Jan. 2026 – May 2026

CSU Channel Islands – Department of Computer Science

Camarillo, CA, under Profs. Abdoolee & Vakilian

- Addressed **LLM output containment** and **adversarial robustness** gaps in production agentic systems by producing two IEEE-format papers spanning experimental design, formal modeling, and applied safety evaluation
- Solved raw-PII exfiltration to external model providers by sole-authoring the **PII Virtualization** paper – designed the mechanism, ran the experimental protocol against a live deployment, wrote the paper end-to-end
- Tackled agentic-safety formalism by co-authoring **ACDS**; owned engineering framing of the POMDP formulation, the adversarial threat taxonomy, and the LANL telemetry evaluation. Defended design trade-offs in weekly research syncs
- Research interests: adversarial robustness of LLM tool-calling, **capability containment**, safety-constrained evaluation, **AI control**

Founder & Sole Engineer

2025 – Present

Traciora – AI Customer Operations Platform

Remote

- Built an AI platform that runs the operational side of a live automotive business – analyzing every customer phone call, handling routine customer operations, and answering from a complete, always-current picture of the company
- Architected a **custom agentic AI harness** (on **LangChain** + **LangGraph's deepagent framework**) with **47 tools across 11 categories**, role-based capability gating, and a rule+LLM **GuardrailAgent** for input/output screening with a complete audit trail
- Built a **daily ensemble speech-to-text pipeline** orchestrating four HuggingFace-sourced neural STT models (Faster-Whisper, OpenAI Whisper, GigaAM, Yandex) with **two-stage LLM transcription restoration**, turning ignored call recordings into per-call recommendations and a rolled-up manager report
- Shipped a **reversible session-scoped PII tokenization layer** (SHA-256 + Fernet + fail-closed leak guard); reduced raw PII exposure to the external LLM provider from **100% to 0%** with zero tool-call degradation – basis of the sole-authored IEEE manuscript
- Engineered an **MLOps / observability stack** on Langfuse capturing every tool call, LLM response, and pipeline stage for offline evaluation, regression testing, and adversarial-prompt review
- Stack: Python 3.11, FastAPI, LangChain, LangGraph (deepagent base), FastMCP, React 18 + TypeScript, SQLite, Docker Compose, Caddy, Langfuse, APScheduler

Data Analytics Specialist

LVCH Auto Tech Center

Mar. 2018 – Mar. 2022

Camarillo, CA

- Developed a **machine learning pipeline** for predictive analytics and customer segmentation, enabling data-driven business intelligence and improving conversion rates
- Engineered an **automated dashboard** with statistical analysis capabilities, providing actionable insights through data visualization and KPI tracking
- Implemented an **A/B testing framework** and feature engineering for campaign performance, improving marketing spend efficiency and customer acquisition

PROJECTS

Kadabra – Systematic Trading Framework | *Conformal Prediction, Time Series, Backtesting* 2023 – Present

- Built an end-to-end framework for researching and deploying systematic crypto strategies: live execution (Binance / Hyperliquid), a multi-test robustness gate (PBO, Deflated Sharpe, MCPT, walk-forward / out-of-sample), and explicit failure recovery
- Applied **conformal prediction** over **100+ signals/minute** across **1TB+ of financial data**; three strategies cleared the gate and trade live (**1.5 Sharpe** in research; **+10.59% / 1.25 Sharpe** live YTD)

AI-Powered Staffing Optimization System | *Gradient Boosting, Feature Engineering, AWS* 2024

- Built a predictive AI agent for Cal Poly Pomona dining services; **1st-place** winning submission at **US AI Camp** (CSU system / AWS / Cal Poly SLO). 94% accuracy on synthetic data; projected 5% labor-cost reduction

TradingView Indicators (Pine Script) | *Signal Processing, Quantitative Features* 2021 – Present

- Authored a curated set of trading indicators and market features in **Pine Script**, published on TradingView and used **daily by 1000+ traders** for discretionary decisions; several reached the **TradingView front page**

TECHNICAL SKILLS

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, HuggingFace Transformers, Ensemble Methods, Conformal Prediction, Walkforward & Out-of-Sample Validation, Experimental Design, Feature Engineering

LLM / Agent Safety: LangChain, LangGraph (deepagent), Tool-Calling Agents, RAG, MCP / FastMCP, GuardrailAgent (input/output screening), PII Detection + Tokenization, Prompt-Injection Defense, Adversarial Threat Modeling, Role-Based Capability Gating

Quant & Statistical Modeling: Time Series Analysis, Distribution-Free Uncertainty, Backtesting, A/B Testing, Statistical Inference, Regression, Classification

Programming Languages: Python, SQL, JavaScript / TypeScript, R, C++, Bash, Unix/Linux

Data, MLOps & Infrastructure: Git, Docker + Docker Compose, FastAPI, AWS, Langfuse, JWT, SQLite, APScheduler, CI/CD, ETL Pipelines, Distributed Systems

Formal Methods & Writing: Constrained POMDPs, NIST Cybersecurity Framework, Staged-Autonomy Taxonomies, Paper Writing (IEEE format), $\text{\LaTeX}$

ACCOMPLISHMENTS

AI & ML Competitions: **1st place US AI Camp** (CSU/AWS), **1st place crypto forecasting** Kaggle competition, **42nd place** DRW crypto forecasting Kaggle competition, **5th place** temperature prediction challenge using conformal prediction methods

Programming Contests: **1st and 2nd place** International Collegiate Programming Contest (ICPC) regionals, among 2-year colleges

Recognition: Advanced Project Management certification; **1000+ users** actively using developed trading indicators